

QC case study – PD scorecard development

You are given a dataset that reports characteristics and delinquency information for 5,960 home equity loans. A home equity loan is a loan where the obligor uses the equity of his or her home as the underlying collateral. The data set has the following characteristics:

- BAD: 1 = applicant defaulted on loan or seriously delinquent; 0 = applicant paid loan
- LOAN: Amount of the loan request
- MORTDUE: Amount due on existing mortgage
- VALUE: Value of current property
- REASON: DebtCon = debt consolidation; HomImp = home improvement
- JOB: Occupational categories
- YOJ: Years at present job
- DEROG: Number of major derogatory reports
- DELINQ: Number of delinquent credit lines
- CLAGE: Age of oldest credit line in months
- NINQ: Number of recent credit inquiries
- CLNO: Number of credit lines
- DEBTINC: Debt-to-income ratio

Analyze the explanatory power of the individual variables (select the short list and perform the univariate analysis), propose an optimal selection of variables, analyze the categorical values (merge or replace by numeric values), and develop an optimal scoring function. Optionally, you may test alternative approaches (e.g. machine learning) to the development of the scoring function. Finally, analyze performance of the scoring function. In order to obtain a non-biased measurement of the scoring function performance, consider a split of the dataset into a training and validation sample, or the cross-validation technique. Summarize your results in a report that starts with an executive summary and then gives details of your development.

Note: You can use any statistical software.